

Homework 3

Instructions:

- Take the acceleration due to gravity to be $g = 9.80 \text{ m/s}^2$
- Express all numerical results to two decimal places

Please submit your answer to the Web Learning by next Tuesday, March 31st.

1. Bullet Fired into a Block

A bullet of mass m is fired into a block of mass M that is initially at rest at the edge of a frictionless table of height h (see Fig. P9.57). The bullet remains in the block, and after impact the block lands a distance d from the bottom of the table. Determine the initial speed of the bullet.

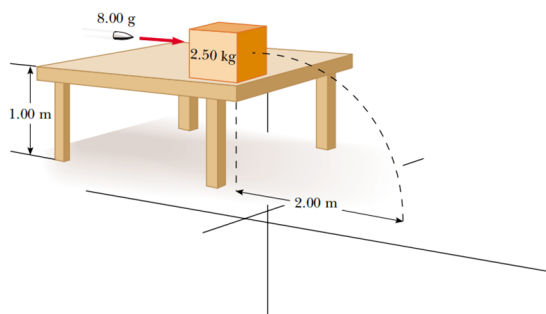


Figure P9.57 Problems 57 and 58.

2. Rocket Propulsion in Deep Space

A rocket for use in deep space is to have the capability of boosting a total load (payload plus rocket frame and engine) of 3.00 metric tons to a speed of 10 000 m/s.

- It has an engine and fuel designed to produce an exhaust speed of 2000 m/s. How much fuel plus oxidizer is required?
- If a different fuel and engine design could give an exhaust speed of 5000 m/s, what amount of fuel and oxidizer would be required for the same task?

3. Falling Chain on a Table

A chain of length L and total mass M is released from rest with its lower end just touching the top of a table, as shown in Figure P9.65a. Find the force exerted by the table on the chain after the chain has fallen through a distance x , as shown in Figure P9.65b. (Assume each link comes to rest the instant it reaches the table.)

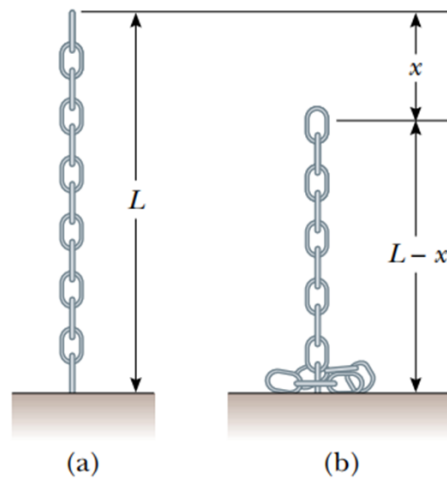


Figure P9.65

4. Center of Mass of a Semicircular Plate

Find the center of mass of a uniform semicircular thin plate.

5. Two Orbiting Astronauts

Two astronauts (see Fig. P11.59), each having a mass M , are connected by a rope of length d having negligible mass. They are isolated in space, orbiting their center of mass at speeds v . Treating the astronauts as particles, calculate (a) the magnitude of the angular momentum of the system. By pulling on the rope, one of the astronauts shortens the distance between them to $d/2$. (b) What is the new angular momentum of the system? (c) What are the astronauts' new speeds?.

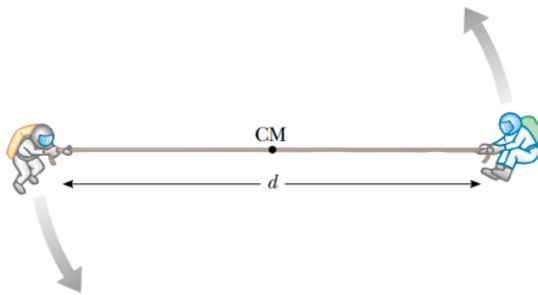


Figure P11.59 Problems 59 and 60.

6. Collision Involving a Rotating Rod

Two particles, each of mass m , are connected by a light rigid rod of length a . In free space, their center of mass is at rest, but the rod rotates about its center of mass with an angular velocity ω . One of the particles on the rod collides with a third stationary particle, also of mass m , and they stick together.

- At the instant just before the collision, where is the center of mass of the three particles? What is the velocity of this center of mass?
- At the instant just before the collision, what is the total angular momentum of the three particles relative to their center of mass? What is it at the instant just after the collision?
- After the collision, what is the angular velocity of the entire system rotating about its center of mass?